Isnad (إسناد) — An Agentic Network Trust Engine

An agentic trust engine built on GSMA Open Gateway CAMARA APIs and the Nokia Network-as-Code platform.

On this revision. This restates the Idea Phase template of 20 August 2026 against the build of 11 September 2026, and supersedes it. The Idea Phase document is deliberately not shipped alongside this one — it described a seven-API proposal and carried claims this repository has since removed, and two versions in one folder only invite a reader to take the wrong one as current. It is retained in the project's working directory and in git history. Claims that had no dated primary source — a cash-on-delivery market share, and categorical statements about what generative AI has broken and what CAMARA is first to do — have been removed rather than restated. Every figure below names the record or command that produces it.

1 · Idea Name & Submission Date

Idea Name	Isnad (إسناد) — Agentic Network Trust Engine
Submission Date	11 September 2026
Phase	Prototype Phase. Revises the Idea Phase entry of 20 August 2026
Code Repository	https://github.com/AhmadShadeed32/isnad
Live demo	https://ahmadshadeed32-isnad-trust-engine.hf.space/judge — no install, no sign-in. Mock evidence by default; an organizer code unlocks bounded real Nokia simulator calls
Run it locally	make setup && make judge → http://127.0.0.1:8010/judge

2 · Submitter Details

Team Name	Isnad
Team Members	Ahmad Shadeed (team lead) and Yousef Al Masri
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Team Size	2
Country	Jordan

3 · Idea Summary

The Problem

Cash on delivery makes both sides of a checkout mistake expensive. A COD order carries no verified commitment at the moment it is placed, so a fake one costs a real delivery run — the van, the driver, the fuel — and a wrongly declined one costs a real sale and, usually, the customer.

The usual defence is a rule: SIM changed recently → decline. A SIM change on its own does not distinguish an account takeover from a customer who replaced a lost phone, because the same signal is produced by both.

A rule that declines on it alone therefore pays twice: once for the fraud it misses on every other path, and once for the legitimate customers it turns away.

Soft proofs of identity are also getting easier to imitate. Deepfakes pass liveness and face checks, voice clones defeat call-centre verification, and one-time passwords remain phishable while adding friction for every honest customer. What a network check contributes that a soft proof cannot is a physical constraint: whether this SIM is in this handset, reachable on this network, in this place, right now.

The Proposed Solution

Isnad is an agentic trust engine. A merchant, wallet or bank calls one API at a moment of risk — signup, checkout, payout, password reset — and asks whether this interaction can be trusted. An investigator agent forms a hypothesis, buys the cheapest useful CAMARA evidence until the answer is justified, and returns ALLOW, CHALLENGE or DECLINE with the full chain of reasoning attached and an Ed25519 signature over it.

The name is literal, not decorative. Isnad (إسناد) is the classical chain of transmission — the method used for over a thousand years to judge whether a report could be trusted, by scrutinising every link that carried it. Every Isnad verdict reproduces that structure: Human → SIM → Device → Location → History, each link a specific CAMARA call, each shown with its own provenance. The internet never got its isnad.

The claim is deliberately narrow, and it is the one the prototype can demonstrate: that a SIM change is worth investigating with network checks before it is acted on, and that the investigation should be inspectable and signed.

Why an investigator agent, not a fixed pipeline

- **Forms a hypothesis from the request context** — for example, a new account at high value paying cash on delivery reads as elevated account-takeover risk.
- **Buys the cheapest useful evidence first**, rather than running every API by default.
- **Escalates only when the belief state justifies the cost** — it calls SIM Swap, Device Swap, Location Verification or Reachability when the hypothesis makes them relevant, not on a schedule.
- **Stops when policy is satisfied** — 49 provider operations against 78 for a run-everything pipeline over the same thirteen authored cases. That is a count of calls, not money.
- **Issues a verdict with the full chain attached** — never a black-box score.

The choice of which CAMARA API to call next is itself the agent's reasoning, which is what the hackathon asks for: CAMARA calls as agent-initiated decision inputs rather than user-triggered buttons. The boundary inside the agent is explicit and visible in the trace — the model chooses what to buy, policy decides what the answer means, and policy can require a check the model skipped. Each verdict records which strategy actually ran, so a fallback to greedy can never be presented as a model run.

Expected Benefits

None of the following is measured. It is the intended shape of the benefit, and it stays a hypothesis until the pilot protocol in docs/MERCHANT_PILOT_PLAN.md produces outcome data — a protocol written before the results are known, and starting in shadow mode so no real customer decision changes first.

- **Merchants and couriers:** fewer fake and refused COD orders, and fewer vans sent to nobody.
- **Banks and wallets:** account-takeover defence at payout and password reset that does not depend on OTPs, selfies or documents.

- **Customers:** a legitimate SIM replacement gets a step-up instead of a decline, and a clean signup gets no OTP at all.
- **Financial inclusion:** thin-file customers with no bank record assessed on SIM tenure and location stability. Most fraud tools shrink a market; this one is designed to grow one.
- **Operators:** a measurable revenue line for CAMARA, and a recurring reason for merchants to consume network capability.

4 · Project Type & GSMA Pillar Alignment

Project Type	AI agent and B2B API platform — a working prototype, deployed and publicly reachable, with a reproducible judge walkthrough
Primary Pillar Alignment	Anti-fraud and trusted digital identity — Open Gateway network APIs consumed as real-time trust signals rather than as user-facing features
Secondary Alignment	Financial inclusion and digital finance enablement — approving thin-file customers on SIM tenure and location stability
Platform	Nokia Network-as-Code — CAMARA APIs against the sandbox and the hosted simulator
Maturity	Prototype complete and deployed. 1,605 checks passed on 11 September 2026 — 1,458 non-browser and 147 browser, covering English and Arabic. 28 recorded Nokia hosted-simulator attempts, failures included. Receipts verify offline and fail on a single changed byte

5 · Theme (One of 7)

Selected Theme	Theme 4 — Secure Fintech, Payments & Anti-Fraud Innovation
Theme APIs	The theme lists SIM Swap, Number Verification and Device Status. Isnad uses all three as core links in the chain, and adds Location Verification, Device Reachability, Device Roaming Status, Number Recycling and Congestion Insights when the agent judges the extra evidence worth its cost
Fit	Isnad is an anti-fraud layer for payments: it protects cash-on-delivery e-commerce and wallet onboarding, defends against account takeover at payout and password reset, answers vishing through Reverse Isnad, and extends payment access to customers with no formal financial record
Adjacent Theme	Theme 1 — Trusted Digital Identity & Cross-Border Verification is the closest neighbour, and the same engine serves it with a different hypothesis. Theme 4 is selected because the wedge, the business case and the demo all lead with payments fraud

6 · API Usage

Eight CAMARA APIs with a working adapter on Nokia Network-as-Code — one more than the seven proposed at the Idea Phase, now including Number Recycling and Congestion Insights. The agent selects which to call at run time; not one of them is bound to a button.

CAMARA API	Role in the evidence chain	Agent trigger	Recorded against Nokia
Number Verification	Consent-based authentication — proves the number belongs to this device. Replaces the OTP.	Always first: the lowest-cost identity check	Sandbox — CONSENT_REQUIRED
SIM Swap	The primary account-takeover signal.	On elevated takeover suspicion	Hosted simulator
Device Swap	The mule-phone signal, and the corroborator that saves customers.	When swap or new-account risk is live	Hosted simulator
Location Verification	Boolean against an address the customer already claimed. Never a coordinate, never a map.	On COD and delivery-address risk	Sandbox
Device Reachability	Bot-farm and burner-phone patterns.	On synthetic-identity suspicion	Sandbox
Device Roaming Status	Impossible travel; doubles as a stability proxy for thin-file customers.	On travel anomalies and inclusion checks	Sandbox
Number Recycling	Was this number reassigned to someone else?	When account age and number history are in doubt	Hosted simulator
Congestion Insights	Network information only, quarantined from the verdict — it never touches the risk score and never waives a check.	When a slow measurement needs an explanation	Hosted simulator

Number Verification's sandbox capture returned CONSENT_REQUIRED, so the reachable and consent-required path is verified while a consented PASS is not. A ninth API, Device Intelligence, has no Nokia adapter at all: its sandbox capture returned EVIDENCE_UNAVAILABLE, so the engine returns unavailable rather than inventing a reputation verdict.

Privacy design

- Location Verification returns a yes/no answer against a claim the customer already made. Isnad never receives or stores a coordinate.
- Number Verification is consent-gated: an opaque single-use state, an explicit authorization screen, then a short-lived token. Tokens are never returned in API responses and never persisted in the chain.
- No raw network signals are retained — only the resulting evidence chain, which is Ed25519-signed and independently verifiable.

What has, and has not, been observed against Nokia

- **Observed:** 14 standalone probe calls to Nokia's hosted simulator on 6 September 2026, recorded one line per attempt with the failures intact — a 422, a 503, and a 400 for an out-of-range reference date.
- **Observed:** 14 further attempts made by the application itself on 9 September — ten locally and four driven from the deployed page. Ten of the 14 have a per-attempt HTTP 200 recorded; for the other four the status was not separately captured, and the record claims only what is provable — each carries a normalized signal sourced nac, which this pipeline produces only from a 2xx whose body parsed. Every run ended in a signed receipt. A probe proves the credential and the contract; only an application run proves the integration.

- **Not observed:** anything at all on a live network, any Nokia congestion subscription or callback delivery, and any physical-handset consent round trip. The account is in Simulator mode throughout, and a signed simulator receipt is still a simulator receipt — the interface says so on every evidence row.

7 · Team

Isnad is built by a two-person engineering team. Both members work across the stack; the areas below describe emphasis rather than a hard split of ownership.

Member	Role
Ahmad Shadeed	Team lead · Engineering — agent and policy design, the CAMARA and Nokia NaC integration, the API surface and the evidence vault
Yousef Al Masri	Engineering, co-builder — shares the technical build across the agent, the API, the provider integration and the test suite

What the team built

Area	What was built
Agent and policy engine	The investigator agent, its hypothesis and belief model in log-odds, and the evidence-cost surface that decides which CAMARA call is worth making next. Gemini is the model planner; greedy is the deterministic fallback
CAMARA / NaC integration	The provider adapter for eight CAMARA APIs, the hosted-simulator and Nokia-contract mock paths, the offline fake operator, and the consent-gated Number Verification flow
API and evidence vault	Verification with idempotent replay, Reverse Isnad, trust sessions with live revocation, and the Ed25519-signed evidence chain that verifies offline and in the browser
Surfaces, tests and docs	The judge, API-key, console, lab and receipt pages in English and Arabic, the 1,605 checks of 11 September including Playwright coverage of both languages, and this submission package

Approved tooling only: Google AI Studio (Gemini) is the single model provider, listed in the Resource & Tooling Guide as a free-tier model API. No second model, and no third-party agent framework.

8 · Current Build Status

- One verification endpoint with Idempotency-Key support, signed-chain retrieval, and offline verification of the exact stored receipt bytes.
- Reverse Isnad — verifying an inbound caller to the customer, so a vishing call can be answered with network evidence.
- Trust with a TTL: sessions keep watching the SIM after approval and revoke a live session on a mid-session swap.
- Consent-gated Number Verification with NaC authorization and callback resume, contract-tested against a local fake operator.
- Five surfaces — /judge, /api-keys, /lab, /console and /docs — with a full RTL Arabic interface. The composed verdict prose is still English, and the page says so rather than implying otherwise.
- A judge-minted merchant API key: behind the same organizer code, /api-keys issues a short-lived key shown once and prints the guide's /v1/verify request filled in, so a judge can call the API from a terminal as a

merchant would. The key drives the deployment's mock provider only, and is refused where the default provider makes billable calls. Checked on the deployed Space on 11 September 2026.

- Judge-selectable evidence source: mock by default, with an organizer code unlocking real Nokia hosted-simulator calls that are capped per run, per session and per deployment hour.
- An Ed25519-signed evidence vault. Receipts verify offline and in browser WebCrypto, and fail on a single changed byte.
- Deployed and publicly reachable, verified on 9 September 2026: anonymous requests answered, both paired real scenarios driven from the deployed page, four Nokia HTTP 200 responses, verified receipts, tamper rejection, and replay with no new provider calls.

Repository: https://github.com/AhmadShadeed32/isnad · Live: https://ahmadshadeed32-isnad-trust-engine.hf.space/judge · Judge guide: submission/JUDGE_GUIDE.md

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